Qiming Liu

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EDUCATION

Bachelor of Science in Data Analytics Expected Graduation: May 2027

Missouri State University | Springfield, MO | GPA: 3.94/4.0

- Relevant Coursework: Database Management, Applied Business Statistics, Data Mining, Machine Learning

EXPERIENCE

Intern — AI Engineer August 2026 — Present

OriLoan Financial Inc.

- Engineered an automated pricing ingestion and reconciliation platform covering 6 mortgage lenders across 5 heterogeneous portal architectures (REST API, replayable HTTP, ASP.NET WebForms, Blazor/SignalR, JSF), replacing manual rate sheet processing with a workflow that validated quotes against each lender's official pricing engine.
- Reverse-engineered the lender's undocumented pricing rules into a deterministic model with no fitted parameters, raising pricing accuracy from 44.6% to 100% across 12,000 tested rate points and resolving a key blocker to launching a consumer rate product.
- Implemented k-fold cross-validation across pricing scenarios and gated all 6 lenders in automated CI regression, ensuring pricing accuracy is continuously verified and protected from production regressions.
- Hardened a public pricing API with rate limits for each caller, input validation and CloudWatch monitoring. Accelerated delivery by directing AI coding agents to build importers, run pricing sweeps at scale and analyze residuals.

Research Jun 2026 – Present

Medicaid Expansion and County-Level Health & Economic Outcomes in Missouri — Spatial Difference-in-Differences

- Investigating whether Missouri's 2021 Medicaid expansion causally improved county-level health and economic outcomes, and whether those gains spill over to neighboring counties, producing the county-level evidence that prior state- and national-level studies lack, to inform state healthcare policy.
- Assembled a panel dataset spanning all 114 Missouri counties from five federal sources, engineering health and economic indicators: uninsured rate, preventable hospitalizations, diabetes mortality, poverty, median income, education, and age structure.
- Specifying a Spatial Difference-in-Differences model with county and year fixed effects to isolate the treatment effect and quantify cross-county spillovers across the state.

PROJECTS & CERTIFICATIONS

Loan Default Prediction — America's Innovate IT Collegiate Conference (AITCC) Analyze IT Challenge | 2nd Place March 2026

- Joined and cleaned three relational tables (70K borrowers, 180K loans, 627K payment records) in Python, handling missing values, duplicates, orphaned foreign keys, and inconsistent formatting to produce a single loan-level modeling dataset.
- Engineered payment behavior features from monthly repayment records including on-time rate and cumulative delinquent days. Correlation analysis showed these far outperformed traditional metrics like credit score and income ($r = 0.76$ vs near-zero).
- Compared Logistic Regression, Random Forest, and XGBoost with stratified validation. Best models achieved AUC 0.96 (classification) and R^2 0.79 (regression). Extended the pipeline into an operational risk monitor with monthly KPI tracking, risk segmentation, a Tableau dashboard, and AI-generated executive reports via Claude API.

Automated Daily Operations Monitoring Pipeline

- Built an automated monitoring pipeline to convert raw operational data into daily KPIs and flag anomalies using a rolling 7 day baseline, improving visibility into daily operational performance.
- Orchestrated scheduled runs via n8n to automate KPI reporting into Google Sheets, achieved 100% automation of the data pipeline from raw ingestion to visualization, eliminating the need for manual data entry.
- Improved monitoring consistency by standardizing daily checks and enabling earlier detection of revenue drops and cancellation spikes, supporting faster and more proactive operational decision making.

IBM Data Analyst Professional Certificate

- Executed end to end analytical workflows using Python, SQL, Excel, AI assisted analysis, and data visualization tools to cleanse, interrogate, and synthesize real world datasets into dashboards and decision ready insights.

LEADERSHIP & ACTIVITIES

Arts & Culture Department Committee Member | Student Government Association

- Organized a university sports performance involving 300 participants, coordinating recruitment, rehearsals and event logistics. Organized six weeks of rehearsals for a dance team of over 20 members that earned Third Prize in a university dance competition.

SKILLS

- Programming Languages: Python, SQL, R
- Data Analysis & Machine Learning: pandas, scikit-learn, XGBoost, Random Forest, Logistic Regression, Feature Engineering, Cross-Validation, Difference-in-Differences, Advanced Excel
- Data Visualization & BI: Tableau, Power BI, Dashboard Development, KPI Reporting
- Engineering & Automation: REST APIs, ETL Pipelines, Git, CI/CD, AWS CloudWatch, n8n, Claude API, AI-Assisted Development